

● SIMULATION PROJECT · ECE / RF ENGINEERING

2x2 Microstrip Patch Antenna Array Design

Designed & Simulated for 2.4 GHz WiFi

$S_{11} = -34$ dB

$VSWR = 1.25$

Gain = 10.11 dBi

$\eta = 90.2\%$

Project Overview

What was designed, and why it matters.

2.4 GHz Operating Frequency	10.11 dBi Directivity / Gain	−34 dB S11 at Resonance
1.25 VSWR at 2.4 GHz	90 MHz Bandwidth (−10 dB)	90.2% Radiation Efficiency

What is a 2×2 Array?

- Four identical microstrip patch elements arranged in a 2-row × 2-column grid.
- Fed equally via a corporate T-junction feed network on FR4 substrate.
- Constructive interference from all 4 patches gives +3 dB gain over a single patch.

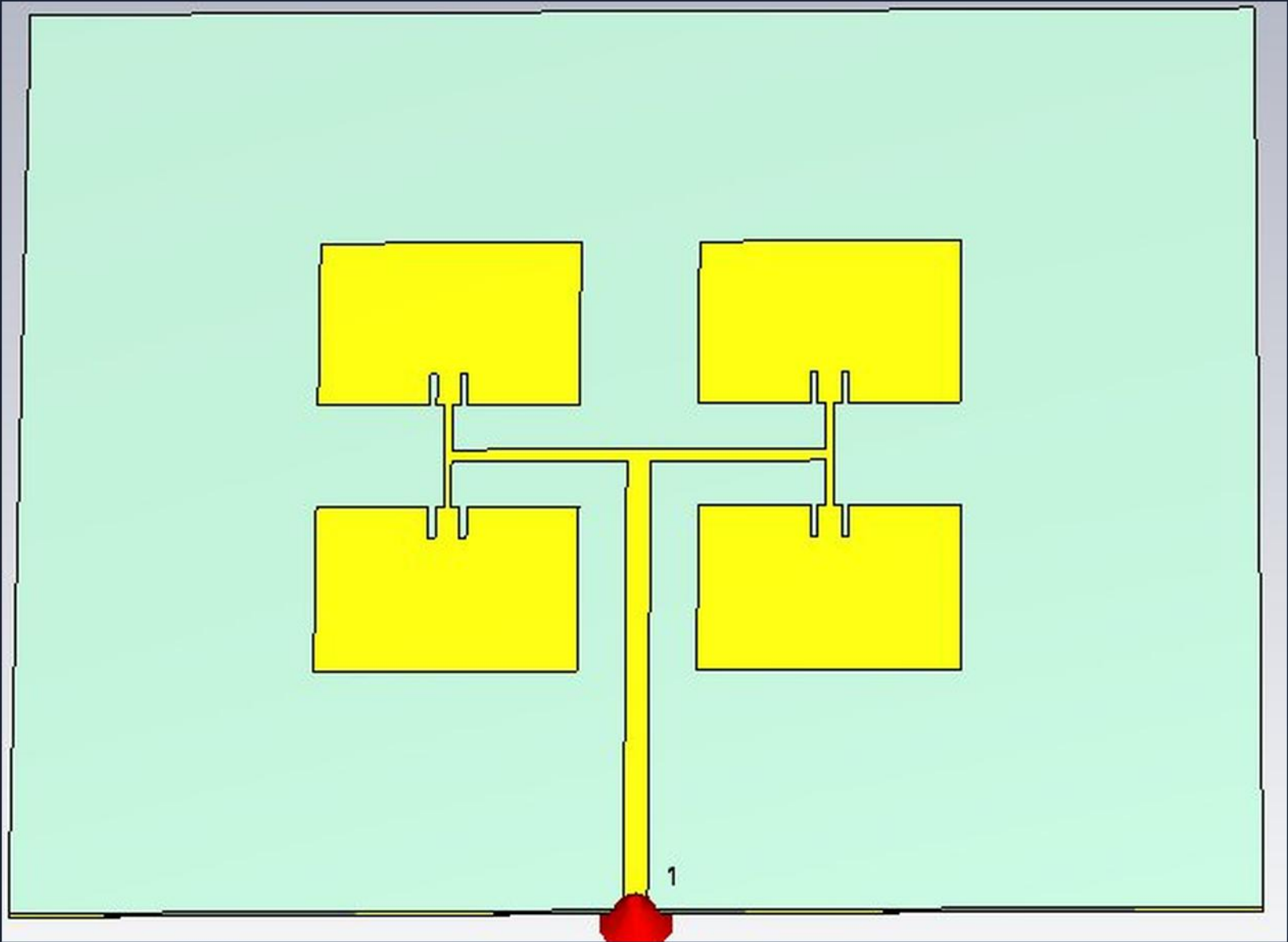
Quick Comparison

	Single Patch	2×2 Array (This Design)
Gain	~7 dBi	10.11 dBi (+3.11 dB)
S11	~−20 dB	−34 dB
VSWR	~2.0	1.25

Antenna Dimensions

All key dimensions calculated & optimised in CST.

Substrate	FR4 ($\epsilon_r = 4.4$, $h = 1.6$ mm)
Patch Length L	39.8 mm
Patch Width W	55 mm
Feed Inset y_0	~8.0 mm
Element Spacing	$0.5\lambda = 62.5$ mm
Ground Plane	260 × 220 mm
Feed Line (50 Ω)	3.1 mm wide
Quarter- λ (70.7 Ω)	1.55 mm wide
Board Thickness	1.6 mm
Copper	0.035 mm (1 oz)



▲ CST Microwave Studio model — 2×2 patch array with corporate T-junction feed on FR4

Patch Dimension Design

Step-by-step equations → optimised in CST

1 Patch Width W

Sized for max radiation efficiency from resonant edge.

$$W = (c / 2f_0) \times \sqrt{2 / (\epsilon_r + 1)}$$
$$W = (3 \times 10^8 / 2 \times 2.4 \times 10^9) \times \sqrt{2 / 5.4} = 0.0625 \times 0.6086$$
$$W = 38 \text{ mm} \rightarrow \text{CST optimised: } W = 55 \text{ mm}$$

W = 55 mm (CST)

2 Effective Dielectric Constant ϵ_{eff}

Fringing fields travel through air AND substrate.

$$\epsilon_{\text{eff}} = (\epsilon_r + 1) / 2 + (\epsilon_r - 1) / 2 \times (1 + 12h / W)^{-\frac{1}{2}}$$
$$\epsilon_{\text{eff}} = 2.7 + 1.7 \times (1.349)^{-\frac{1}{2}} = 2.7 + 1.463$$
$$\epsilon_{\text{eff}} = 4.163$$

$\epsilon_{\text{eff}} = 4.163$

3 Fringing Extension ΔL

E-field extends beyond edges — subtract $2\Delta L$.

$$\Delta L = 0.412h \times [(\epsilon_{\text{eff}} + 0.3)(W/h + 0.264)] / [(\epsilon_{\text{eff}} - 0.258)(W/h + 0.8)]$$
$$\Delta L = 0.659 \times 154.59 / 137.36 = 0.741 \text{ mm}$$

$\Delta L = 0.741 \text{ mm}$

4 Patch Length L

$$L_{\text{eff}} = c / (2 \times f_0 \times \sqrt{\epsilon_{\text{eff}}}) = 3 \times 10^8 / (4.8 \times 10^9 \times 2.040) = 30.6 \text{ mm}$$
$$L = 30.6 - 2 \times 0.741 = 29.1 \text{ mm} \rightarrow \text{CST optimised: } L = 39.8 \text{ mm}$$

L = 39.8 mm (CST)

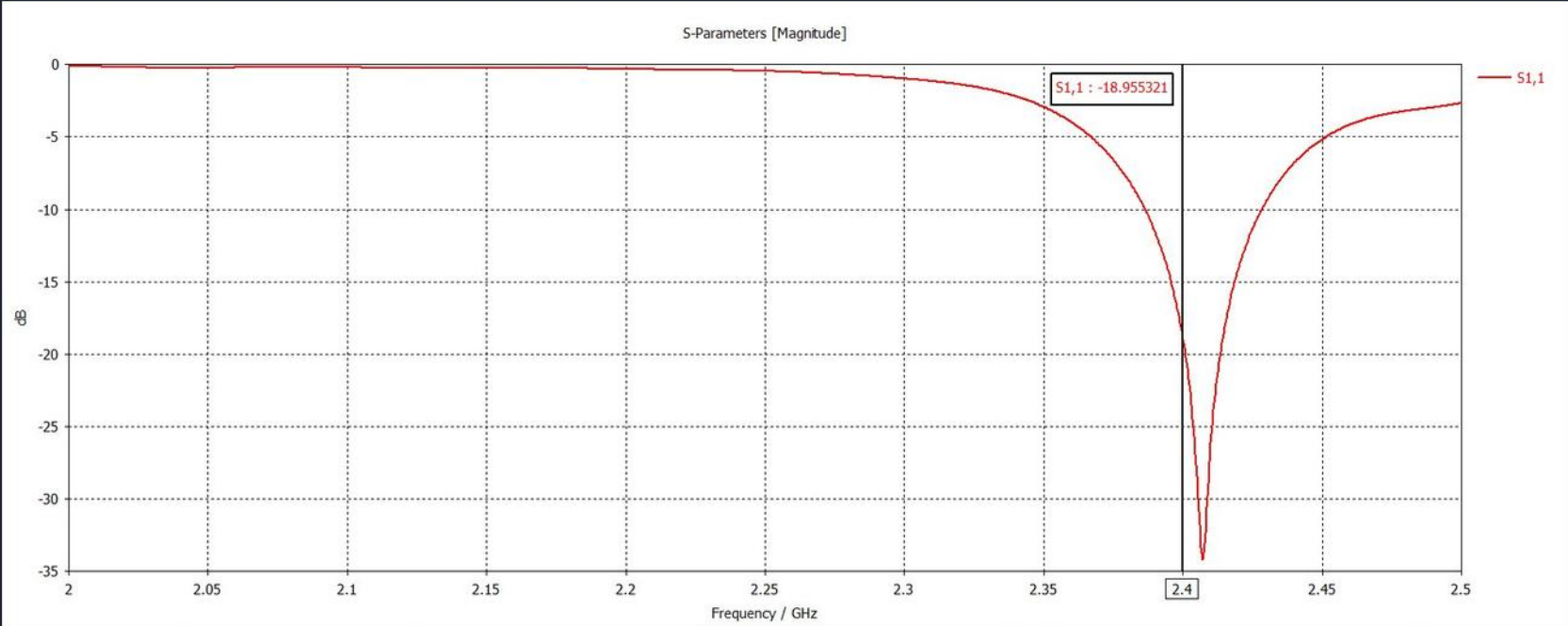
5 Inset Feed y_0 — 200 Ω edge → 50 Ω match

$$Z_{\text{in}}(y_0) = Z_{\text{edge}} \times \cos^2(\pi \times y_0 / L)$$
$$y_0 = (L / \pi) \times \arccos(\sqrt{50 / Z_{\text{edge}}}) \rightarrow y_0 = 8.0 \text{ mm}$$

$y_0 = 8 \text{ mm} \rightarrow \text{perfect } 50 \Omega \text{ match}$

S11 Return Loss

−34 dB at resonance · Marker at 2.4 GHz = −18.96 dB



▲ S11 magnitude (dB) vs frequency — deep resonance dip at 2.40 GHz, marker reads −18.96 dB

2.40 GHz

Resonance

−34 dB

Min S11

−18.96 dB

@ 2.4 GHz Marker

< −10 dB

Match Threshold

Parameter	Value	Status
Resonance Frequency	2.40 GHz	✓ Perfect
S11 at resonance (dip)	−34 dB	✓ Excellent
S11 at 2.4 GHz marker	−18.96 dB	✓ Good
Lower −10 dB freq.	~2.355 GHz	✓
Upper −10 dB freq.	~2.445 GHz	✓
Bandwidth	90 MHz	✓ Full WiFi coverage

S11 & VSWR Math

Full reflection coefficient derivation and power analysis

$S_{11} = 20 \times \log_{10} |\Gamma|$ Rule: < -10 dB = Acceptable | < -15 dB = Good | < -20 dB = Excellent

1 $|\Gamma|$ from resonance $S_{11} = -34$ dB

$|\Gamma| = 10^{(-34/20)} = 10^{(-1.7)} = 0.0200$
Reflected power = $(0.020)^2 \times 100 = 0.04\%$ – almost zero!

Only 0.04% reflected at resonance (-34 dB)

2 VSWR at 2.4 GHz (marker $S_{11} = -18.96$ dB)

$|\Gamma| = 10^{(-18.96/20)} = 10^{(-0.948)} = 0.1124$
 $VSWR = (1 + 0.1124)/(1 - 0.1124) = 1.1124/0.8876 = 1.2532 \approx 1.25 \checkmark$

VSWR = 1.25 ✓ Excellent (limit < 2.0)

3 Power Budget

Reflected = $(0.1124)^2 \times 100 = 1.26\%$
Radiated = $100 - 1.26 = 98.74\%$
Safe TX power: 7–8 W continuous | 12–15 W burst

98.74% of input power is radiated

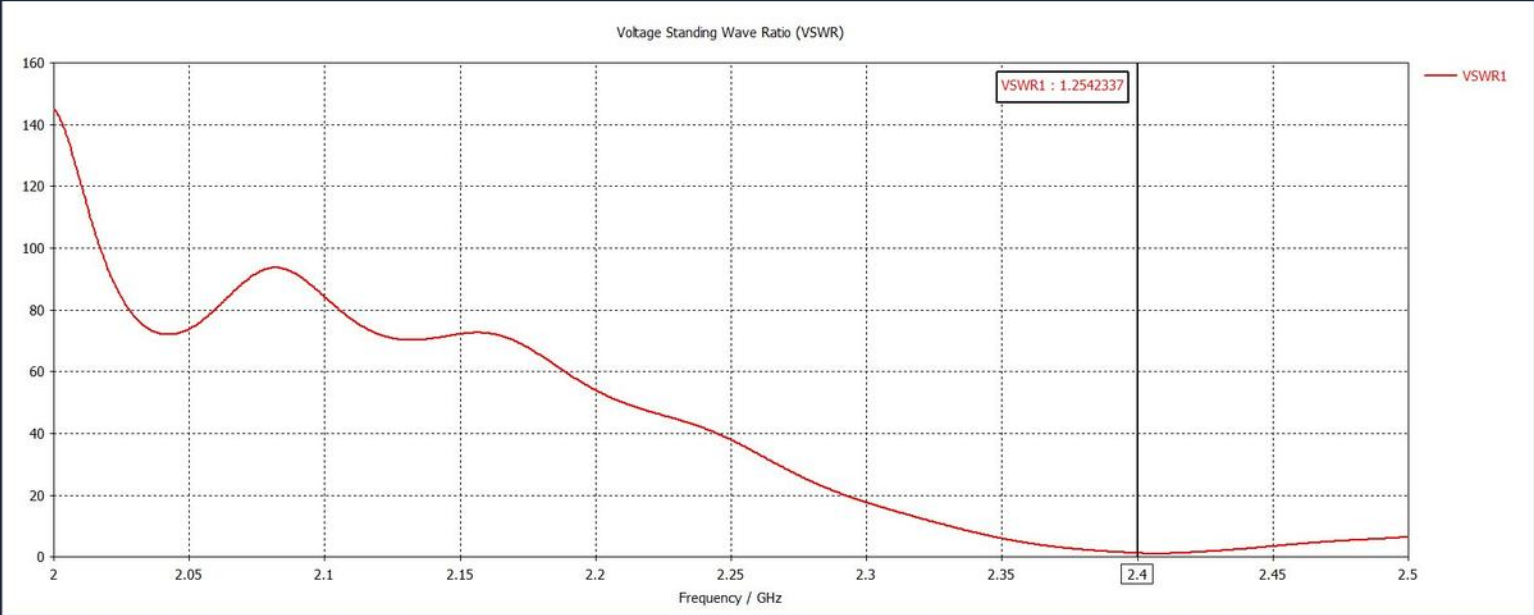
4 Cross-check: $|\Gamma|$ from VSWR

$\Gamma = (VSWR - 1)/(VSWR + 1) = 0.25/2.25 = 0.1111$
 $S_{11} = 20 \times \log_{10}(0.1111) = -19.08$ dB matches -18.96 dB ✓

Cross-check passed ✓

Voltage Standing Wave Ratio

VSWR = 1.25 at 2.4 GHz · Near-perfect impedance match



▲ VSWR vs frequency — drops to 1.25 at 2.4 GHz (marker). Far below 2.0 acceptable threshold.

1.25

VSWR @ 2.4 GHz

1.0

Ideal VSWR

< 2.0

Acceptable Limit

98.7%

Power Delivered

VSWR Formula & Verification

$\Gamma = (VSWR - 1) / (VSWR + 1) = (1.25 - 1) / (1.25 + 1) = 0.25 / 2.25$
 $|\Gamma| = 0.1111$
 $S_{11} = 20 \times \log_{10}(0.1111) = -19.08 \text{ dB} \quad \checkmark \text{ (matches marker } -18.96 \text{ dB)}$
Reflected power = $|\Gamma|^2 \times 100 = 1.23\%$
Transmitted power = 98.77% → nearly all input power is radiated

Bandwidth & Power Handling

–10 dB impedance bandwidth + copper thermal limit

1 Bandwidth (–10 dB criterion)

$$f_1 \text{ (lower -10 dB)} = 2.355 \text{ GHz} \quad , \quad f_2 \text{ (upper -10 dB)} = 2.445 \text{ GHz}$$
$$BW = f_2 - f_1 = 0.090 \text{ GHz} = 90 \text{ MHz}$$
$$\% BW = (90/2400) \times 100 = 3.75\%$$

BW = 90 MHz (3.75%) — full WiFi 802.11 b/g/n ✓

WiFi Standard	Freq Range	Required BW	Our BW	Status
802.11b	2.400–2.483 GHz	83 MHz	90 MHz	✓ Covered
802.11g	2.400–2.483 GHz	83 MHz	90 MHz	✓ Covered
802.11n	2.400–2.483 GHz	83 MHz	90 MHz	✓ Covered

2 Dielectric Breakdown (theoretical ceiling)

$E \propto \sqrt{P} \rightarrow$ find P at FR4 breakdown field 20 MV/m.

$$E_{sim} = 12,593.9 \text{ V/m @ 1 W (CST output)}$$
$$P_{max} = 1 \times (20,000,000/12,593.9)^2 = 2.52 \text{ MW (theoretical)}$$

Dielectric ceiling = 2.52 MW (thermal limit is real constraint)

3 Copper Thermal Limit (actual safe power)

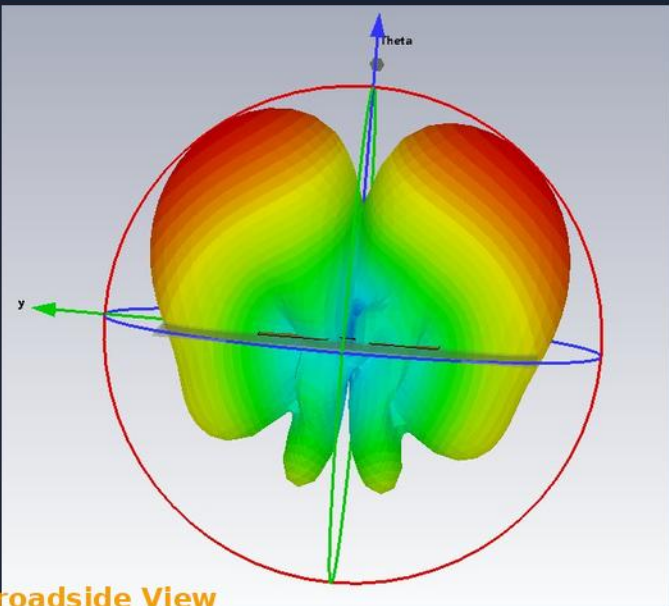
Skin effect + RF derating factor 2.5.

$$P_{copper} = I^2 \times R = 0.7656 \times 25 = 19.14 \text{ W (DC)}$$
$$RF \text{ derated} = 19.14/2.5 = 7.66 \text{ W}$$

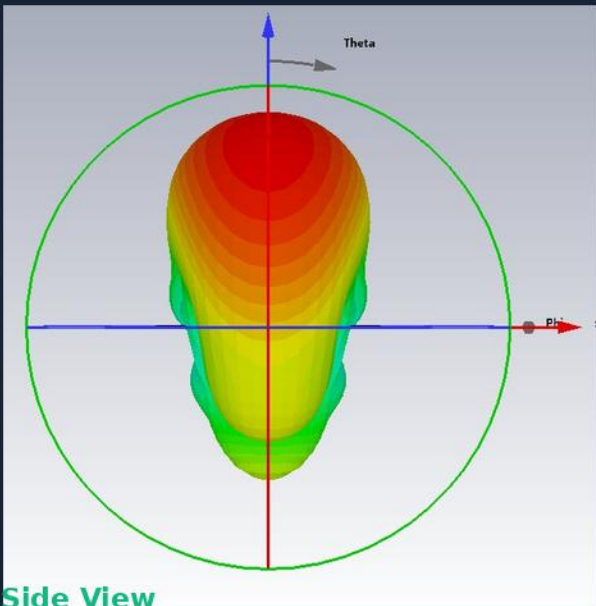
Safe: 7–8 W continuous | 12–15 W burst

Radiation Pattern & Efficiency

10.11 dBi Directivity · 90.2% Radiation Efficiency · ~40° HPBW



Front / Broadside View



Elevation / Side View

farfield (f=2.4) [1]	
Type	Farfield
Approximation	enabled (kR >> 1)
Component	Abs
Output	Directivity
Frequency	2.4 GHz
Rad. Effic.	-0.4384 dB
Tot. Effic.	-0.4940 dB
Dir.	10.11 dBi

Efficiency Breakdown

Directivity
10.11 dBi

Rad. Efficiency
-0.4384 dB → 90.22%

Tot. Efficiency
-0.4940 dB → 89.06%

Realized Gain

- Radiation is broadside — perpendicular to patch surface, ideal for ceiling-mounted WiFi APs.
- Array narrows beam from ~80° (single patch) to ~40° HPBW — concentrating energy forward.
- Ground plane suppresses back radiation — all energy pushed to front hemisphere.

Gain & Efficiency

CST dB output → %, realized gain & array improvement

Directivity = beam focus (100% eff). Gain = D × η_{rad}. Realized Gain also accounts for mismatch loss.

1 Radiation Efficiency dB → %

CST reports −0.4384 dB — convert to physical percentage.

η_{rad} = 10^{^(−0.4384/10)} = 10^{^(−0.04384)} = 0.9022 = 90.22%
Loss = 9.78% − heat in FR4 dielectric + copper resistance

η_{rad} = 90.22%

2 Total Efficiency dB → %

η_{tot} = 10^{^(−0.4940/10)} = 10^{^(−0.0494)} = 0.8907 = 89.07%
This includes both radiation loss AND impedance mismatch loss.

η_{tot} = 89.07%

3 Realized Gain

G_{realized} = Directivity + Tot. Efficiency (dB)
G_{realized} = 10.11 + (−0.4940) = 9.62 dBi

Realized Gain = 9.62 dBi

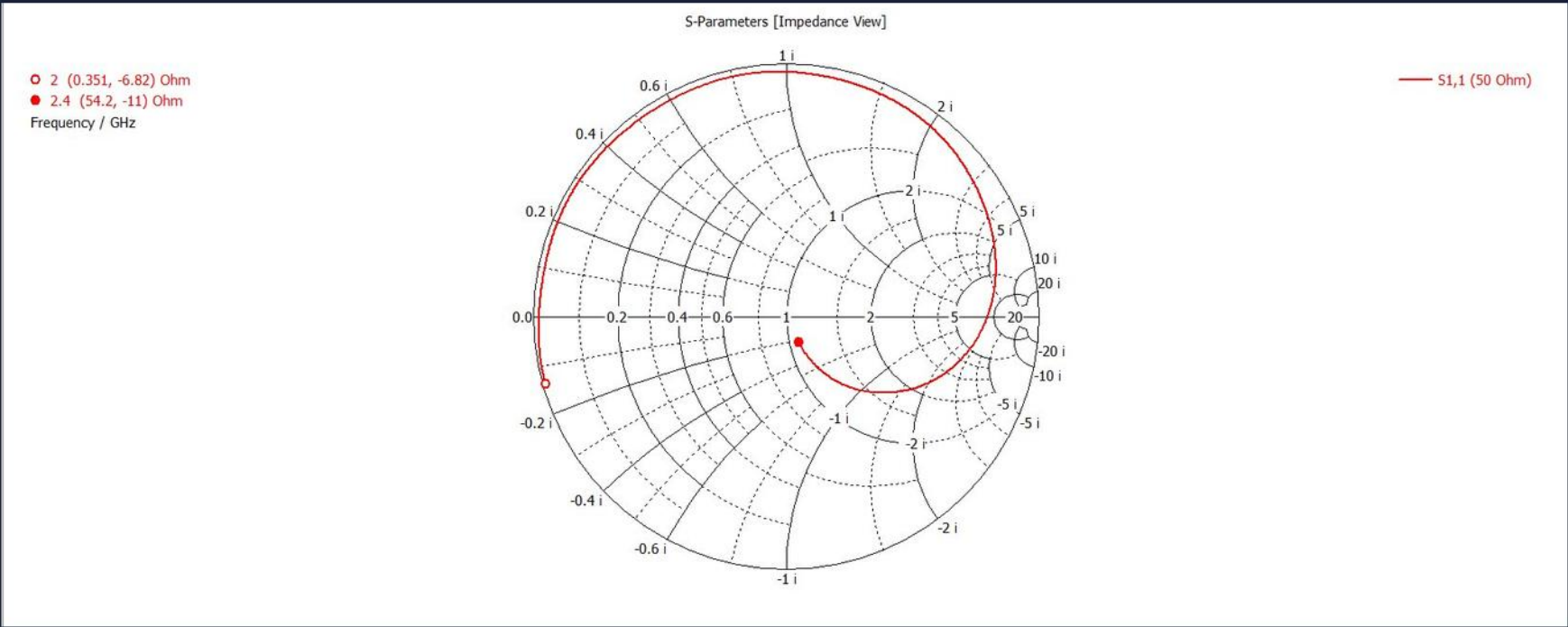
4 Gain Improvement over Single Patch

Single patch gain = ~7.0 dBi
Array gain = 10.11 dBi
Improvement = +3.11 dB = 2.04× power in forward direction

+3.11 dB gain (2× power boost over single patch)

Smith Chart Analysis

$Z = 54.2 - j11 \, \Omega$ at 2.4 GHz · Very close to 50 Ω centre



▲ Smith Chart — trace spirals from 2.0 GHz (open circle, far left) to 2.4 GHz (filled dot near centre)

Frequency	Impedance	Assessment
2.0 GHz (open circle)	0.351 − j6.82 Ω	Far from 50 Ω — severe mismatch
2.4 GHz (filled dot)	54.2 − j11 Ω	✓ Near 50 Ω — excellent match

Analytical Verification of S11 from Impedance

$\Gamma = (Z_L - Z_0) / (Z_L + Z_0)$ where $Z_L = 54.2 - j11 \, \Omega$, $Z_0 = 50 \, \Omega$

$|\text{num}| = |(54.2-50) - j11| = |4.2 - j11| = \sqrt{(17.64+121)} = \sqrt{138.64} = 11.78$

$|\text{den}| = |(54.2+50) - j11| = |104.2 - j11| = \sqrt{(10857.6+121)} = \sqrt{10978.6} = 104.78$

$|\Gamma| = 11.78 / 104.78 = 0.1124 \rightarrow S_{11} = 20 \times \log_{10}(0.1124) = -18.99 \, \text{dB} \quad \checkmark$

Smith Chart Verification

Analytical $|\Gamma|$ from $54.2-j11\ \Omega$ — matching simulated S11

Smith Chart centre = $50+j0\ \Omega$. At 2.4 GHz: $Z_L = 54.2-j11\ \Omega$. We compute $|\Gamma|$ analytically and verify vs simulated $S_{11} = -18.96\ \text{dB}$.

1 $\Gamma = (Z_L - Z_0) / (Z_L + Z_0)$ $Z_0 = 50\ \Omega$

Numerator: $(54.2-j11) - 50 = 4.2 - j11$
Denominator: $(54.2-j11) + 50 = 104.2 - j11$

2 Magnitude $|a+jb| = \sqrt{a^2+b^2}$

$|num| = \sqrt{4.2^2 + 11^2} = \sqrt{17.64 + 121} = \sqrt{138.64} = 11.775$
 $|den| = \sqrt{104.2^2 + 11^2} = \sqrt{10857.6 + 121} = \sqrt{10978.6} = 104.78$

3 Final $|\Gamma|$ and S11

$|\Gamma| = 11.775 / 104.78 = 0.1124$
 $S_{11} = 20 \times \log_{10}(0.1124) = -18.99\ \text{dB}$
Simulated S11 at 2.4 GHz = $-18.96\ \text{dB}$ → matches! ✓

S11 = -18.99 dB (calc) vs -18.96 dB (CST) ✓ Verified

Frequency	Impedance (Ω)	$ \Gamma $	S11 (dB)	Quality
2.0 GHz	$0.351 - j6.82$	~ 0.99	-0.09	Severe mismatch
2.4 GHz	$54.2 - j11$	0.1124	-18.99	✓ Excellent

Single Patch vs 2x2 Array

How much does the array improve over a single element?

❑ Single Patch Element	
Gain	~7 dBi
S11	~-20 dB
VSWR	~2.0
Bandwidth	~80 MHz
HPBW	~80°
Efficiency	~85%
Board Size	55×41 mm
Feed	Simple direct
Complexity	Low

■ 2x2 Array ← THIS DESIGN	
Gain	10.11 dBi (+3.11 dB)
S11	-34 dB
VSWR	1.25
Bandwidth	90 MHz
HPBW	~40° (narrower)
Efficiency	90.2%
Board Size	260×220 mm
Feed	Corporate T-junction
Complexity	Moderate

⚡ Array Gain Calculation

$G_{array} = G_{single} + 10 \times \log_{10}(N) = 7.0 + 10 \times \log_{10}(4) = 7.0 + 6.02 = 13.02 \text{ dBi (ideal)}$
Simulated: 10.11 dBi — diff 2.91 dB due to mutual coupling & substrate losses

Complete Results Summary

All simulation parameters at a glance.

#	Parameter	Value	Status
1	Operating Frequency	2.4 GHz	✓
2	S11 at resonance (min)	−34 dB	✓ Excellent
3	S11 at 2.4 GHz marker	−18.96 dB	✓ Good
4	VSWR at 2.4 GHz	1.25	✓ Excellent
5	Bandwidth (−10 dB)	90 MHz	✓
6	% Bandwidth	3.75%	✓
7	WiFi band coverage	2.355 – 2.445 GHz	✓ Full
8	Directivity	10.11 dBi	✓
9	Radiation Efficiency	90.22%	✓
10	Total Efficiency	89.06%	✓
11	Realized Gain	~9.62 dBi	✓
12	Impedance at 2.4 GHz	54.2 – j11 Ω	✓ Near 50 Ω
13	Γ Reflection Coeff.	0.1124	✓
14	Power reflected	1.26%	✓
15	Gain vs single patch	+3.11 dB improvement	✓

Design Successful

All performance targets met or exceeded.

✓ **S11 = −34 dB at resonance** · Excellent matching

✓ **VSWR = 1.25** · Near-perfect impedance match

✓ **90 MHz bandwidth** · Full WiFi 802.11 b/g/n coverage

✓ **10.11 dBi directivity** · +3.11 dB vs single patch

✓ **90.2% radiation efficiency** · minimal substrate loss

✓ **Ready for real-world fabrication on FR4 PCB**

CST Microwave Studio

FR4 PCB

Antenna Array

2.4 GHz WiFi

IEEE 802.11

RF Engineering

Microstrip

EM Simulation

ECE Dept.